Hardware connections

This board uses a **CH32V003** with an **ILI9341** TFT over **SPI1**. **MISO is not connected** (write-only path). Pins below match the **CH32V003 TFT module** wiring you use.

East Rising **ER-TFTM028-4** (optional exact module)

If your panel is the **ER-TFTM028-4** breakout (*Datasheet V2.0*, buydisplay.com / East Rising), it matches this project:

Item	Spec (from module DS)
Size / resolution	**2.8″**, **240 × RGB × 320**
Controller	**ILI9341**
Bus options	8080 parallel (8/9/16/18), **3-wire SPI**, **4-wire SPI**, RGB — this firmware uses **4-wire SPI** (separate **CS**, **SCK**, **SDI**, **D/C**)
TE	**JP1 pin 22** — tearing-effect **output** from the panel (frame sync); tie to **PD0** + EXTI as in your schematic
CS	**JP1 pin 23** `LCD_/CS`, active low
SCK	**JP1 pin 24** `D/C(SCL)` acts as **serial clock** in 4-wire SPI
D/C	**JP1 pin 25** `/WR(D/C)` = **command/data** select in 4-line serial
SDI	**JP1 pin 27** `LCD_SDI` (MOSI)
RESET	**JP1 pin 21** `/RESET`, active low (module may have RC reset; wire if you drive reset from MCU)
Backlight	**JP1 pin 29** `BL_ON/OFF` — high on / low off (or power mode per jumper **J9–J12**; see module §4.3)
SDO	**JP1 pin 28** — not needed for write-only

Critical — solder jumpers: The module ships for **8080 16-bit parallel by default** (§4.3). For this firmware you must set **4-wire SPI**: **J2, J3, J4, J5 short** and **J1, J6, J7, J8 open**, with the resistor stuffing called out in the same table. If jumpers stay on parallel mode, SPI bit-banging from the MCU will not drive the ILI9341 correctly.

Power: Module supports **3.3 V or 5 V** `VDD` (JP / strap per §4.3); **I/O** is **3.3 V logic** (`VDDIO`). Match **CH32V003** at **3.3 V** and follow East Rising notes for `VDD` strap vs backlight jumpers.

Module pinout (your design — MCU side)

TFT / module signal	CH32V003 pin	Direction	GPIO mode	Description
VCC	3.3 V supply	—	—	Power (**3.3 V only** — do **not** use 5 V)
GND	GND	—	—	Common ground
Crystal (X1)	**PA1 / PA2**	IN/OUT	Analog	24 MHz external crystal → 48 MHz system clock
CS	**PC1**	OUT	Push-pull	Chip-select (**active LOW**)
RESET	**PD4**	OUT	Push-pull	Hardware reset (**active LOW**)
DC/RS	**PD3**	OUT	Push-pull	Data/command (**H** = data, **L** = command)
SDI / MOSI	**PC6**	OUT	AF-PP	**SPI1 MOSI**
SCK	**PC5**	OUT	AF-PP	**SPI1** clock
LED (backlight)	**PD2**	OUT	AF-PP	Backlight **PWM** (**TIM1_CH1**)
TE (tearing effect)	**PD0**	IN	Input (floating or pull-up per panel)	Panel frame sync → **EXTI** (see below)
SDO / MISO	**NC**	—	—	Not connected

The firmware drives **PC1** (CS), **PD3** (D/C), and **PD4** (RESET) via GPIO **BSHR**; **PC5** / **PC6** are **SPI1** alternate-function push-pull. **PD2** backlight uses **TIM1_CH1 PWM** (see below). **PD0** is the **TE** input; enable **EXTI** on **PD0** and NVIC for `exti7_0_irqhandler` .

Backlight PWM (PD2 / TIM1_CH1)

Implemented in `src/hw.rs` :

API	Role
`backlight_on()`	Full brightness — calls `backlight_set_duty(1023)`
`backlight_set_duty(duty)`	Set duty cycle; values above **1023** are clamped

Timer setup: TIM1 CH1 on **PD2**, PSC = 0, `ATRLR` = **1023** → PWM frequency ≈ **48 MHz / 1024 ≈ 46.9 kHz** (10-bit resolution).

Duty range: `0` = off, **`1023` = maximum** (`BACKLIGHT_PWM_MAX`). The compare register is `TIM1->CH1CVR` .

`backlight_set_duty(n)`	Approx. brightness
`0`	Off
`256`	~25 % (256 / 1023) — **not** the maximum
`512`	~50 %
`1023`	**100 %** (full on)

At the end of `ili9341::init()` , the firmware calls `backlight_on()` so the panel starts at full brightness. To dim:

```
hw::backlight_set_duty(512); // ~50 %
```

Peripheral addresses (as coded)

Peripheral	Base / register	Purpose
SPI1	`0x40013000`	8-bit TX to the panel (`SR` poll / `DR` write)
GPIOC BSHR	`0x40011010`	**CS** on **PC1** (atomic set/reset)
GPIOD BSHR	`0x40011410`	**D/C** on **PD3**, **RST** on **PD4**
TIM1	`0x40012C00`	Backlight **PWM** on **PD2** (`CH1CVR` duty)

Register bases follow the usual CH32V00x map: GPIO port A `0x40010800` , B `0x40010C00` , C `0x40011000` , D `0x40011400` ; **BSHR** is at offset **+0x10**.

Tearing effect (TE) / EXTI

Signal	MCU pin	Notes
TE	**PD0**	Input from the ILI9341/module **TE** pad; route through **EXTI** so `exti7_0_irqhandler` runs on the panel's tearing/blanking edge (configure rising vs falling per your panel and scan direction).

The symbol `exti7_0_irqhandler` matches the WCH **EXTI line 0–7** interrupt. Map **PD0** to the correct **EXTI line** and source in the CH32V003 reference manual (AFIO / EXTI pin mux), enable the interrupt in **NVIC**, and set trigger polarity to match **TE**.

In this repository, EXTI pending clear in the ISR is still a TODO ("platform specific"). Add the correct clear for the line wired to **PD0** if the IRQ sticks or re-enters incorrectly.

WCH datasheet (CH32V003DS0) alignment

Official **CH32V003** datasheet *CH32V003DS0* (e.g. V1.8) matches this board as follows (pin names from **Chapter 2**, **CH32V003F4P6** pinout):

Your net	Datasheet pin functions (abridged)
CS on **PC1**	`PC1` / **NSS** (SPI NSS), plus I2C/timer remap options — use as GPIO push-pull CS if NSS not used in hardware mode.
SCK / **MOSI**	**PC5** = **SCK**; **PC6** = **MOSI** (same SPI).
D/C PD3	`PD3` — general-purpose I/O / alternate functions (`A4` , `UCTS` , etc.).
RST PD4	`PD4` — includes `UCK` , timer / SPI-related remaps; fine as GPIO reset.
BL PWM PD2	`PD2` — **TIM1_CH1** (`T1CH1`) matches backlight on **TIM1_CH1**.
TE PD0	`PD0` — **TIM1_CH1N** and remappable `SDA` / `UTX` ; configure as **input + EXTI** for TE.

Your net	Datasheet pin functions (abridged)
Crystal PA1 / **PA2**	**OSCI** / **OSCO** for **HSE** 4–25 MHz (your 24 MHz crystal fits).

Device memory from the same document: **2 KB SRAM**, **16 KB CodeFlash** — consistent with `memory.x`.

EXTI (§1.4.8): eight external interrupt lines; multiple port pins can be tied to one EXTI source — configure **PD0** and the EXTI mux in the **RM** (datasheet is summary only).

Electrical notes

- CH32V003 I/O is **3.3 V**; the table assumes a 3.3 V module supply.
- One-way SPI to the ILI9341 is normal with **MISO NC**.